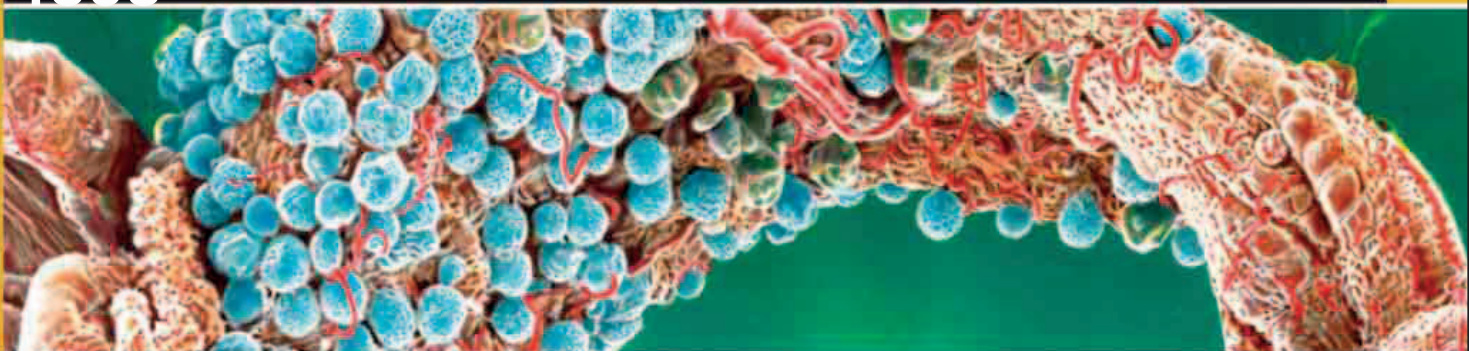




Marconi formed his Wireless Telegraph and Signal company in 1897.



Malaria "oocysts" (blue) in the mosquito stomach lining, first observed by Ronald Ross, burst into cells that infect the insect's salivary glands—and thereby, its bite.

of medical technology, such as the electrocardiogram used for monitoring heartbeat.

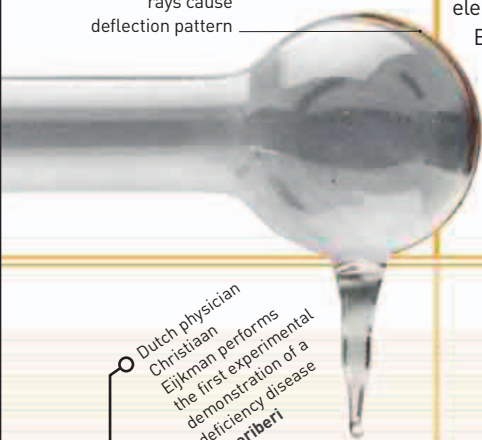
In August, German chemist **Felix Hoffmann** (1868–1946), working at the Bayer pharmaceutical laboratories, produced a painkilling substance called acetylsalicylic acid. It was modeled on a related ingredient derived from certain medicinal plants, such as willow and meadowsweet, which had been known since Ancient Greece. The new painkiller was later marketed as **aspirin** (see 1899).

Another medical breakthrough was made by German physician **Paul Ehrlich**. He developed the **side-chain theory**, which explained how the **immune system** could attack specific infections. It remains the basis of immunological theory to this day.

Cathode ray tube

Glass vacuum tubes proved useful to scientists in the discovery of X-rays and electrons. Rays pass along the tube and the pattern formed at the end shows they are negatively charged particles (electrons).

rays cause deflection pattern



AFTER HENRI BECQUEREL

DISCOVERED RADIOACTIVITY,

Polish–French physicist **Marie Curie** and her husband, French physicist **Pierre Curie** (1859–1906), embarked on a lifelong career studying radioactivity at the Becquerel Laboratory. Becquerel had found that pure uranium emissions could cause air to conduct electricity. The Curies discovered that a uranium ore—called pitchblende—was 300 times stronger in this respect, and deduced that a new element present in the ore must be responsible. They named the element **polonium**, after Marie's native country Poland, and coined the term **radioactive** at the same time. Later that year, the Curies discovered another radioactive element, **radium**, and managed to purify quantities of both for further study.

By the 1890s, scientists had discovered two **unreactive noble gases**, helium and argon, but William Ramsay suggested that gaps in the periodic table and laboratory analysis of air pointed to the existence of other elements. In 1898, working with British chemist Morris Travers (1872–1961), he discovered three more noble gases: **krypton**, **xenon**, and **neon**.

In July, the Annual Meeting of the British Medical Association reported

1,000 nm

BACTERIUM

Virus 20–40 nm

Size differential between virus and bacterium

Viruses are measured in nanometers (nm). They lack the cellular structure of bacteria, being just particles of protein and genetic material.

on key discoveries about a deadly disease—malaria. British physician **Ronald Ross** (1857–1932), working in India, had proved that mosquitoes spread the **malaria parasite** through their bite. The previous year—after painstakingly dissecting mosquito guts—he had found malaria parasites lodged in the stomach walls of these insects.

His work demonstrated that the life cycles of the malaria parasite (see 1893–94) and certain kinds of mosquitoes were linked.

Meanwhile, Dutch biologist **Martinus Beijerinck** (1851–1931) made another breakthrough. He found that a plant disease called **tobacco mosaic** could be spread even when infected plant extract was passed through a filter that held back bacteria. He deduced that the contagious particles were smaller than bacteria, and called them viruses. Beijerinck's tobacco mosaic virus would not be isolated until the 1930s.

In Germany, physicist **Wilhelm Wien** (1864–1928), experimenting with the positively charged rays produced in certain types of vacuum tubes, laid the foundations of a new area of analytical science, **mass spectrometry**. This was a technique used to determine the make-up of molecules by vaporizing them into ions (charged

particles). Wien invented a way of separating the different kinds of ions in electromagnetic fields according to their mass and charge. The accuracy of mass spectrometry has resulted in it being used in fields as diverse as the medical testing of blood and urine to analyzing atmospheric samples in space exploration.



MARIE CURIE (1867–1934)

Born in Poland, Marie married Pierre Curie in France in 1895. The couple shared the Nobel Prize in 1903 with Henri Becquerel for their work on radioactivity. Marie received a second Nobel Prize in 1911 for the discovery of polonium and radium. She donated all her medals to the World War II effort. She died of leukemia caused by radiation exposure.

“NOTHING IN LIFE IS TO BE FEARED, IT IS ONLY TO BE UNDERSTOOD. NOW IS THE TIME TO UNDERSTAND MORE, SO WE... FEAR LESS”

Marie Curie, Polish–French physicist

Dutch physician Christiaan Eijkman performs the first experimental demonstration of a deficiency disease—beriberi

American chemist John Abel describes the extraction of a **derived form** of the hormone epinephrine

February 17 Marie and Pierre Curie find that **uranium ore** is more **radioactive** than pure uranium

April Italian physician Camillo Golgi describes the cellular structure later called **Golgi apparatus**

May–July William Ramsay and Morris Travers isolate the elements **krypton**, **neon**, and **xenon**

July 8 Ronald Ross traces **malaria parasites** to the salivary glands of **mosquitoes**

July 13 Pierre and Marie Curie show that the high **radioactivity of uranium ore** comes from a new element, later called **polonium**

Wilhelm Wien describes streams of **ionized gases**, later called **protons**, and establishes basis of **mass spectrometry**

December 26 Pierre and Marie Curie discover a new radioactive element, later called **radium**

December German microbiologist Carl Benda coins the term **mitochondria**